energywave.ai • Seed Round — technical documents, evidence status and operating requirements
16 October 2026 · EnergyWave Final

CONFIDENTIAL — FORECAST — FOR DISCUSSION · NO MEASURED ENERGYWAVE PERFORMANCE EXISTS · LEAD PATH: SWEET SOLID-STATE

What a technical reviewer receives with the seed round. Every document, its status, and what it does and does not establish.

The technical hub for the seed round and the companion to the Seed Round Structure page, the Seed Deck and the Seed One-Pager: the technical material a seed investor's reviewer is given, what each item establishes, and the standard every later result is held to. It states no measured EnergyWave performance, because none exists; it says what would have to be measured, and replicated, for Tranche B to be released.

SCOPE STATEMENT — READ BEFORE ANY DOCUMENT IN THIS FOLDER

> No measured EnergyWave hardware performance evidence has been reviewed for this package. Every number in an EnergyWave-authored document is labelled by origin — *assumption · calculation · proposed target · company measurement · attributed published result* — and no EnergyWave document carries a company measurement yet. Figures cited from published research are attributed to their authors and are not EnergyWave results. Every claim the company examines must answer two questions: **what is the net energy source, and where is the colder sink?** A refinement adopted here: a prepared, stored or driven state is itself an answer to the first question and is counted as an input; a colder sink is the test for a heat-engine claim, and the release of a stored potential is accounted as storage, never as generation.

WHAT THE SEED FUNDS, TECHNICALLY

Tranche	Technical deliverable	Evidence that releases the next tranche
A released at close	Principal investigator hired (a condition of the tranche). Thread C: the artifact-exclusion bench commissioned on known-null samples; its error budget published before any result is claimed (Error Budget page). Thread B: the pre-registered replication of Moddel et al. 2021 — cavity / no-cavity twins from the same wafer, blinded measurement, sign and magnitude against cavity thickness. Thread A — the lead path since 15 October 2026: the founder's solid-state module characterized under exactly the same protocol once the inventor-defined inputs are on record at kickoff (Thread A plan).	Gate-1 report. Pass: repeatable output above the fully characterized noise and artifact floor at ≥5σ in blinded runs, surviving every control. Kill: nothing above the floor, or output explained by a known source — published with data. Either outcome is a result; only the first releases Tranche B.
B escrowed	Gate 2 — closed-system calorimetry; mechanism discrimination against known-source models (finite stored-energy depletion, open-system environmental transduction, metrology phase skew); independent replication at a second laboratory contracted before Gate 2 begins.	Gate-2 report: net output whose closed-system energy balance cannot be closed by known sources, replicated by the second laboratory. Otherwise the effect is retired publicly.

Tranche	Technical deliverable	Evidence that releases the next tranche
C escrowed	Gate 3 — engineering characterization, lifetime and stability, third-party validation under observation; the anti-tamper “paperweight test”.	The first point at which EnergyWave would ever state a performance number. Scaling, lifetime or stability failure ends the program having answered the question.

Tranche amounts, the seed amount and the instrument are set with counsel; the projected sizing in the Seed Deck and One-Pager comes from the forecast model (v1.2, Base scenario) on assumptions to be validated by finance; no SAFE executed, no outside funding raised.

ENERGY ACCOUNTING — THE STANDARD EVERY THREAD IS HELD TO

> **$E_{\text{all inputs}} = E_{\text{useful load}} + E_{\text{other leaving}} + \Delta U_{\text{device}}$**, where U includes everything stored in capacitors, batteries, magnetic and strain states and internal thermal reservoirs. Preparation, drive, control, cooling and reset energy are counted once each; a reset already inside a full-cycle measurement is not counted twice. DC input is ∫v(t)i(t)dt, never average V × average I; probe deskew in a high-Q circuit can turn a true +50 mW dissipation into a false −75 mW “generation” reading at 1 MHz. A null result bounds the tested configuration at its demonstrated sensitivity — it does not identify an origin. A voltage recovery, a resonant amplification ratio, a short-circuit current trace or a single discharge is an observation, not a useful-load energy measurement.

THE TECHNICAL DOCUMENT SET

Document	What it establishes — and does not
Research Program and Gates Technical Package	The research question, the three-gate protocol with pass / kill criteria, the energy-accounting standard, threads A / B / C under one protocol with no dependency asserted, the eight commitments. Sweet solid-state is the lead path (15 October 2026); the bench order and the measurement lead are set at kickoff.
Thread A Characterization Plan Technical Package	The first testable question for the founder’s module; hypotheses H-A1 stored-energy depletion, H-A2 environmental transduction, H-A3 metrology phase skew; off-the-shelf baseline proposals; the four-step unconditioned characterization; pre-registered pass / kill / ambiguous rule. The specimen, magnetization, winding axis, conditioning procedure and the founder’s notes are inventor-defined and specified at kickoff.
Measurement Platform Error Budget Technical Package	Thread C for external review: the measurand (output power into a defined load), the error-budget table with every term “to be measured” until commissioning, the commissioning sequence on known-null samples, calorimetry, sign-off. The ≤ 0.14 pW floor is a *proposed target* inherited from the prospectus, ratified at commissioning; effect size, resolution and the leads are set at commissioning.
Humanoid Power-Budget Specification Technical Package	Capital-readiness objective #2: method, attributed platform benchmarks, the six-row subsystem worksheet and the specification outputs ($\bar{P}$, P_{peak}, ramp, mass, volume, heat rejection). Platform inputs are set at the design review. It defines the question a power source would have to answer; it asserts nothing about any EnergyWave device.
Physical Architecture & Baseline Model v1.3 (5 October 2026) Technical Package · 6 pp	Working briefing: reservoir inventory, energy accounting, metrology, the three conditioning hypotheses, an unconditioned baseline characterization before any conditioning experiment. Reference; its §6 corporate table is superseded by the Corporate, IP and Legal Status page.

Document	What it establishes — and does not
Claims Discipline every package	The rules every document obeys — two-question filter, evidence labels, no device figures, publish both ways.
xTech Search 10 white paper (approved); Energy at the Point of Use v2.1 Government Package	Thread B as proposed to the Army: an independent, pre-registered evaluation of the MIM-diode / Casimir-cavity claim (Moddel et al. 2021) under pre-declared criteria — run on the thread C measurement platform, which is the metrology, not the claim; the submitted cover states TRL 2. A submission is not an award and not a validation. The v2.1 paper limits “unbuilt theoretical estimates” to kilowatt-scale extraction figures and states Moddel’s picowatt-scale result as disputed and not independently replicated.
Compartmentalization & Anti-Tamper Posture v0.1 (12 September 2026) Government Package	How a positive result would be protected (CPI / AT). A v0.1 working memo; its §8 questions are being worked with counsel.
Literature cross-check of the eight archive papers (14 October 2026)	Three independent readings of the archive literature, reconciled: legitimate physics and useful history; no repeatable vacuum-powered engine, no validated ferrite mechanism, no measured EnergyWave performance. Secondary evidence, not replication; the findings are carried into the table below.

WHAT THE EVIDENCE ESTABLISHES TODAY

Claim or source	Status	Basis, as the sources state it
A repeatable EnergyWave output of any kind		None measured. No specimen selected, no hardware ordered, no result. Thread A — the lead path since 15 Oct 2026 — has its specimen, magnetization, winding axis and conditioning procedure inventor-defined and placed on record at kickoff.
Moddel et al. 2021, optical-cavity-induced current (MIM diode under a Casimir cavity)		*Attributed published result.* A maximum of 1.4 pW inferred from the I–V curve for one device geometry; repeatability across more than 1,000 devices in 21 batches in the same laboratory. The reported four-hour trace is short-circuit current, not metered useful-load energy. Vacuum excitation is the authors’ proposed interpretation. Maclay 2024 argues the cavity model predicts the wrong current direction and the wrong cavity-size behaviour — a critique of mechanism, not a replication. No independent replication establishing vacuum origin and a complete sustained net-output system was located in the reviewers’ bounded search.
Forward 1984, charged foliated conductors		A proposed rechargeable Casimir battery. Re-expansion of the plates explicitly costs energy from the bidirectional supply (p. 3). A finite discharge with a paid reset: the cycle is non-positive, not a repeating net-power engine. Carried as a one-shot store in Prospectus §6.5.
Puthoff & Little (JBIS 55, 2002; arXiv 2010); Puthoff, Phys. Rev. D 35 (1987); Davis et al., STAIF 2006		Exploratory theory and a proposed experimental program. The 1987 result is an equilibrium balance for a restricted circular orbit (footnote 17 limits it), not surplus power for a load. The 2006 review reports an inconclusive cavity / synchrotron experiment and a switchable-mirror estimate that assumes negligible switching energy; it reports no practicable demonstration as of 2006, which is not a measured universal zero. The local two-author Puthoff & Little record is distinct from the withdrawn three-author record.

Claim or source	Status	Basis, as the sources state it
The Casimir force		Adjacent science (Lamoreaux 1997, torsion-pendulum force measurement). The force is conservative: whatever work it does closing the plates must be paid back to open them. A measured force is not a generator.
Best-case single-use Casimir stack vs. lithium-ion		≈ 2 J/kg against ~9×10⁵ J/kg for the lithium-ion cell a humanoid already carries (source note, 5 October 2026). Said here before an investor's physicist friend says it.
Cook 2002; Utkin 2012; Nair 2021; a ChatGPT conversation export of 2024		Not for any claim. Cook p. 122 reports no demonstrated excess output among ~30 devices; p. 298 orders giga and tera wrongly and compares power to energy. Utkin pp. 54–61 contain preliminary bench observations without a closed energy inventory; a control without the shunt diodes gives a similar trace. Nair Eq. 17 takes the square root of an energy density and keeps energy-density units. The transcript's p. 20 formula adds terms of incompatible dimensions. Preserved as history; cited for nothing.

DRAFTED AND WAITING ON INPUTS; STILL TO WRITE

Item	Owner	When
Thread A characterization plan — drafted; specimen, magnetization, winding axis, conditioning procedure, the founder's notes and prior measurements specified at kickoff	The program lead with Chip Story	Before the first Gate-1 run
Humanoid power-budget specification — drafted; platform, mass and bay volume, duty profile, heat rejection and the six subsystem rows set at the design review	The program lead	Capital-readiness objective #2
Measurement-platform error budget — drafted; the thread A effect size, the calorimetric resolution and the leads set at commissioning	Thread C lead / PI	Commissioning
Pre-registration template, blinding procedure and open-data pipeline specification — to write	PI	Before the first Gate-1 run
Second-laboratory replication terms, contracted before Gate 2 begins — to write	PI with counsel	Gate 2
Scientific advisory board charter with veto over any performance statement; at least one declared critic — to write	Ian McLean	Before Tranche A

OPERATING REQUIREMENTS — FACILITIES, BENCH, STAFFING (QUOTE SCHEDULE IN PREPARATION)

This folder also serves as the technical seed hub — laboratory and workbench, an office near UC Irvine, staged staffing and every other seed requirement, with evidence. The planned office is in the UCI area near the National Fuel Cell Research Center; no lease, facility access, sponsored research, partnership, endorsement or institutional agreement with UCI, the NFCRC or any company there is yet in place, and the plan implies none. A source-backed operating-requirements and quote schedule is in preparation; when received it is placed beside this page and the table below is filled from it. Until then no facility, bench, headcount or cost figure is stated here beyond the forecast model's.

Schedule section	What it carries	Evidence rule
Facilities	Office and laboratory in the UCI area near the NFCRC (planned): candidate spaces, size, services (power, HVAC, vibration, shielding), lease terms, dates.	Dated quote or listing, vendor named.
Minimum bench (thread C)	The artifact-exclusion bench as a parts list: shielded isothermal enclosure, calorimetry, lock-in electrical measurement to the picowatt level, probes and deskew calibration, blinded sample handling, data pipeline — the Error Budget page's terms, costed.	Dated vendor quote per line; a proposed target until purchased.
Staged staffing	Principal investigator at Tranche A; technician; second-laboratory contract at Gate 2; advisory board. Roles and start month by tranche. Current core team Ian McLean and Chip Story; Jack Sarfatti a team member (theoretical physicist); Alan Crunkleton an advisor; Mike Wentz a team member (role to be announced); Arath Acosta, Design Engineer from the Seed Round (planned).	Role described; agreements set with counsel.
Budget evidence	Each line of the forecast model's UseOfFunds that the schedule validates or replaces, with the quote behind it.	Forecast assumptions until validated.
Milestones	Facility ready, bench commissioned on known-null samples, error budget published, first pre-registered run — each tied to a gate month.	Dates are *proposed targets* until the PI signs them.

The technical set is reviewed by independent readers before each edition of these materials; findings are incorporated at source and the documents re-issued.

Basis: the Seed Round Structure, Research Program, Thread A Plan, Error Budget and Power-Budget pages; the 13 October 2026 deck, slides 4–7, 9; the Technical Prospectus v1 §§4, 6.5–6.6; the Physical Architecture & Baseline Model v1.3 §§1–3; the xTech papers; the Compartmentalization & Anti-Tamper Posture v0.1; the literature cross-check of 14 October 2026; the decisions of 15 October 2026.

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